# MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
*(A constituent unit of MAHE, Manipal)*

| DEPARTMENT OF AERONAUTICAL AND AUTOMOBILE ENGINEERING | | | |
|---|---|---|---|
| **END SEMESTER EXAMINATION** | | **NOVEMBER, 2024** | |
| **Level** | **B.Tech** | **Curriculum** | **2022** |
| **Program Name** | **Aeronautical Engineering** | **Course Type** | **C** |
| **Course Name** | **Introduction to Aerospace Engineering** | **Course Code** | **AAE 2125** |
| **Semester** | **III** | **Max Marks** | **50** |
| **Date** | **29-11-2024** | **Duration** | **180 mins** |

**Instructions:**

- **Answer ALL the questions.**
- **Data not provided may be suitably assumed**

| Q. No. | Question | Marks | NBA | | IET | | BL |
|---|---|---|---|---|---|---|---|
| | | | CO | PO | CLO | AHEP4 LO | |
| **1A** | Analyze the primary function of the Hypothetical Model of Atmosphere, also referred to as the International Standard Atmosphere (ISA), in aerospace vehicle design, performance assessment, and testing. Compare and contrast its characteristics with those of the actual atmospheric conditions. Using the International Standard Atmosphere (ISA) model, calculate the expected temperature at an altitude of 5 km, applying the standard mean sea-level temperature and the established lapse rate. | **(05)** | 01 | 1, 2 | **01** | **C1, C2** | **04** |
| **1B** | Explain the following concepts in your own words:<br>a. Boundary Layer Thickness<br>b. Displacement Thickness<br>c. Momentum Thickness | **(03)** | 03 | 1, 4, 5 | **03** | **C1, C2** | **02** |
| **1C** | Analyze and calculate the free-stream Mach number for a full-scale aircraft traveling at 50 m/s at an altitude where the speed of sound is 330 m/s. Based on the resulting Mach number, evaluate the implications of compressibility effects on the aircraft's aerodynamic performance. | **(02)** | 01 | 1, 2 | **01** | **C1, C2** | **04** |
| **2A** | Design a diffuser and nozzle configuration that addresses the following flow conditions: | **(05)** | 04 | 1, 4, 5 | **04** | **C1, C2** | **04** |

<table>
<tr><td></td><td>a. Mach number (M) is less than 1,<br>b. Mach number (M) is equal to 1, and<br>c. Mach number (M) exceeds 1.</td><td></td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td>2B</td><td>How do the different parts of the NACA 4-digit airfoil series nomenclature, such as in 'NACA 2412,' impact the aerodynamic properties of the airfoil? Explain the significance of each digit in terms of airfoil shape and performance.</td><td>(03)</td><td>01</td><td>1, 2</td><td>01</td><td>C1, C2</td><td>03</td></tr>
<tr><td>2C</td><td>In what ways does the Coanda effect enable the generation of anti-torque in a NOTAR (NO TAil Rotor) helicopter?</td><td>(02)</td><td>01</td><td>1, 2</td><td>01</td><td>C1, C2</td><td>03</td></tr>
<tr><td>3A</td><td>How does the design choice of using engines with larger inlet areas and lower thrust per unit mass flow enhance the efficiency of transport aircraft operating at lower speeds? Analyze the relationship between the ratio of exhaust velocity to flight velocity $(u_e/u_0)$ approaching 1 and its effect on increasing propulsive efficiency, $(\eta_{prop})$.</td><td>(05)</td><td>04</td><td>1, 4, 5</td><td>04</td><td>C1, C2</td><td>04</td></tr>
<tr><td>3B</td><td>From the structural elements provided - ribs, bulkheads, stringers, longerons, and spars - determine which components are best suited for the construction of the wings, vertical tail, and fuselage, and justify your choices based on their structural functions.</td><td>(03)</td><td>02</td><td>1, 4, 5</td><td>02</td><td>C1, C2</td><td>03</td></tr>
<tr><td>3C</td><td>Which pressure component remains constant along a streamline in an incompressible, inviscid, steady flow? How can this be determined?</td><td>(02)</td><td>03</td><td>1, 4, 5</td><td>03</td><td>C1, C2</td><td>02</td></tr>
<tr><td>4A</td><td>Identify the effects on the aircraft's orientation or control surfaces that correspond to each of the following aircraft movements or control inputs:
<table>
<tr><td>(1)</td><td>Positive pitching</td><td>(a)</td><td>Right wing down</td></tr>
<tr><td>(2)</td><td>Positive Y-axis</td><td>(b)</td><td>Right wing retreating</td></tr>
<tr><td>(3)</td><td>Positive rolling</td><td>(c)</td><td>Elevator down</td></tr>
<tr><td>(4)</td><td>Positive yawing</td><td>(d)</td><td>Left wing down</td></tr>
<tr><td>(5)</td><td>Negative rolling</td><td>(e)</td><td>Towards right wing</td></tr>
<tr><td>(6)</td><td>Negative yawing</td><td>(f)</td><td>Negative yawing</td></tr>
<tr><td>(7)</td><td>Positive sideslip</td><td>(g)</td><td>Elevator up</td></tr>
<tr><td>(8)</td><td>Negative pitching</td><td>(h)</td><td>Right wing advancing</td></tr>
</table></td><td>(04)</td><td>04</td><td>1, 4, 5</td><td>04</td><td>C1, C2</td><td>03</td></tr>
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|---|---|---|---|---|---|---|---|
| **4B** | Compare and contrast the ideal and real Brayton cycles by analyzing their representations on P-V and T-S diagrams. | **(03)** | 04 | 1, 4, 5 | **04** | **C1, C2** | **03** |
| **4C** | Explain the static stability criterion for rolling, pitching, and yawing motions? | **(03)** | 04 | 1, 4, 5 | **04** | **C1, C2** | **02** |
| **5A** | Describe how eccentricity, semi-major axis, and energy interact to define the shape and characteristics of each type of conic section (circle, ellipse, parabola, and hyperbola). Explain how these relationships influence the types of orbits possible for celestial bodies or satellites. | **(04)** | 05 | 1, 4, 5 | **05** | **C1, C2** | **03** |
| **5B** | Analyze the concept of Lagrangian points in orbital mechanics, examining their significance in the context of gravitational interactions between celestial bodies. Differentiate the conditions that contribute to the stability or instability of each Lagrangian point, and evaluate one practical application of positioning a satellite at a stable or unstable Lagrangian point. | **(03)** | 05 | 1, 4, 5 | **05** | **C1, C2** | **04** |
| **5C** | Explain the characteristics of a geostationary orbit and discuss why this orbit is particularly useful for communication satellites. | **(03)** | 05 | 1, 4, 5 | **05** | **C1, C2** | **02** |

**MARK DISTRIBUTION (CO-wise & BTL-wise)**

| CO / CLO | Total Marks |
|---|---|
| **1** | ***12*** |
| **2** | ***03*** |
| **3** | ***05*** |
| **4** | ***20*** |
| **5** | ***10*** |

| BL | Total Marks |
|---|---|
| **2** | **11** |
| **3** | **19** |
| **4** | **20** |
| **5** | **---** |